Explainable AI for Academic Integrity Verification

Catching AI-written essays fairly, and checking if the student actually understands

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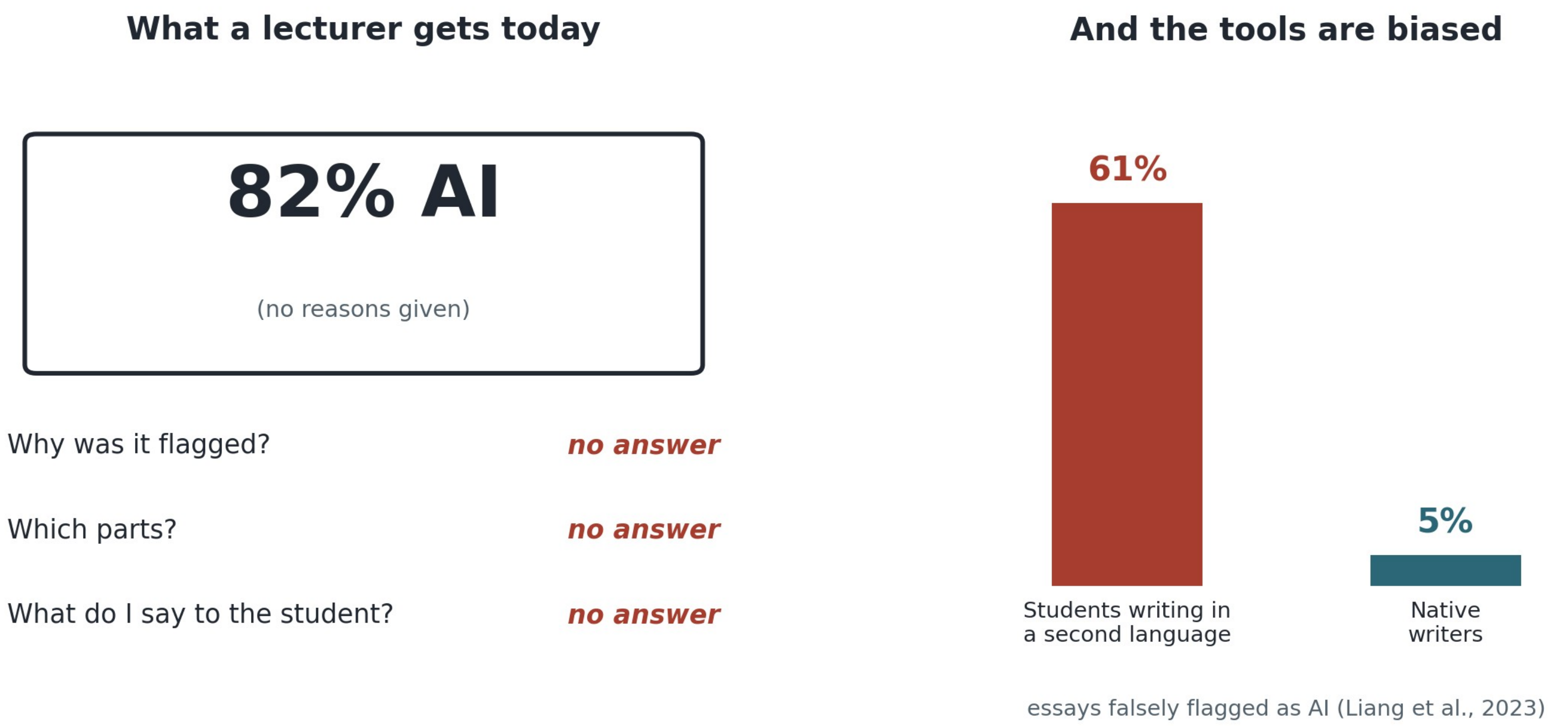
The problem, in plain terms

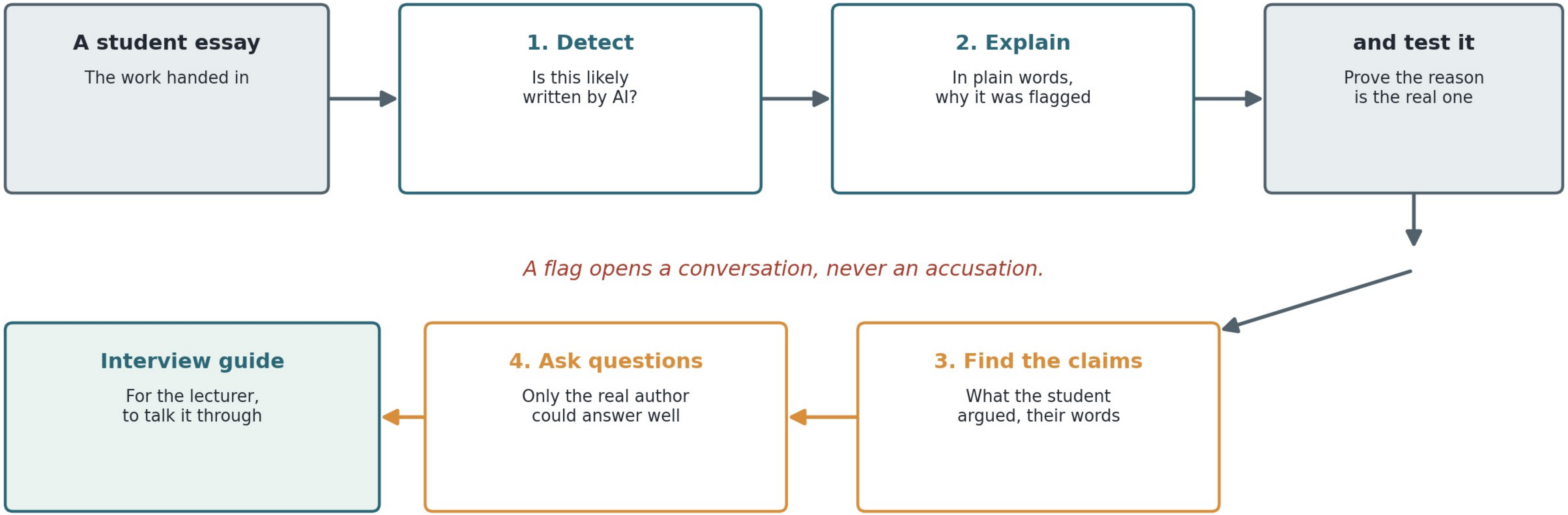
Students can get an AI essay in seconds, and the popular detector tools answer with a bare number a lecturer cannot defend. They are also biased against students writing in a second language. And even a correct number answers the wrong question: HOW the text was made, not whether the STUDENT understands it.



A number nobody can defend, biased against the wrong students, answering how the text was made instead of whether the student understands it.

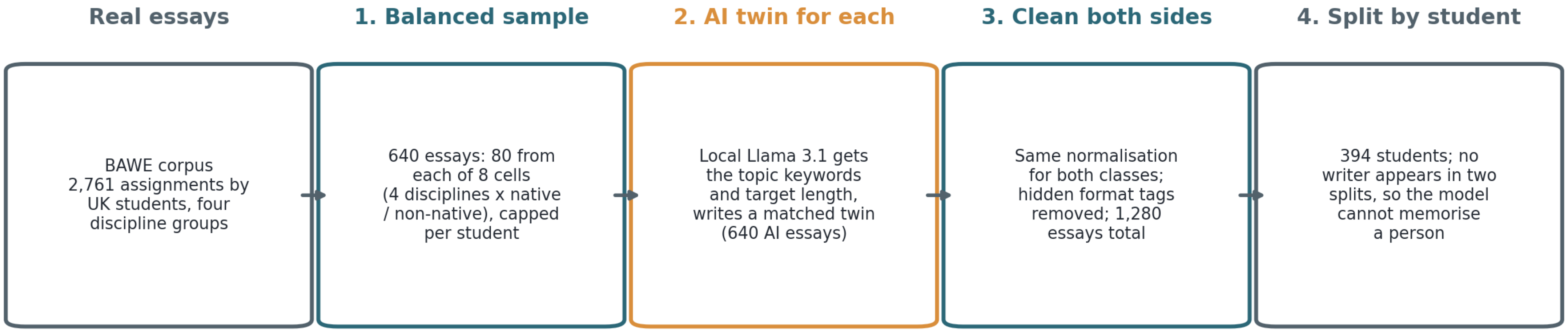
My idea, in one picture

Do not just accuse. (1) Detect if it is likely AI. (2) Explain the flag in plain words, and prove the reason is real. (3) Find the student's own claims (the sentences where they make an argument). (4) Write questions only the true author could answer. Then hand the lecturer an interview guide. A flag starts a conversation, it does not end one.



Building the dataset: every step closes a shortcut

How the 1,280-essay corpus was made (source: the BAWE corpus, Alsop and Nesi, 2009). Sampling is balanced across four disciplines and across native and non-native writers, every AI essay is written to its human twin's topic and length by a local model, both classes get the same cleaning, and no student's writing appears in more than one split.

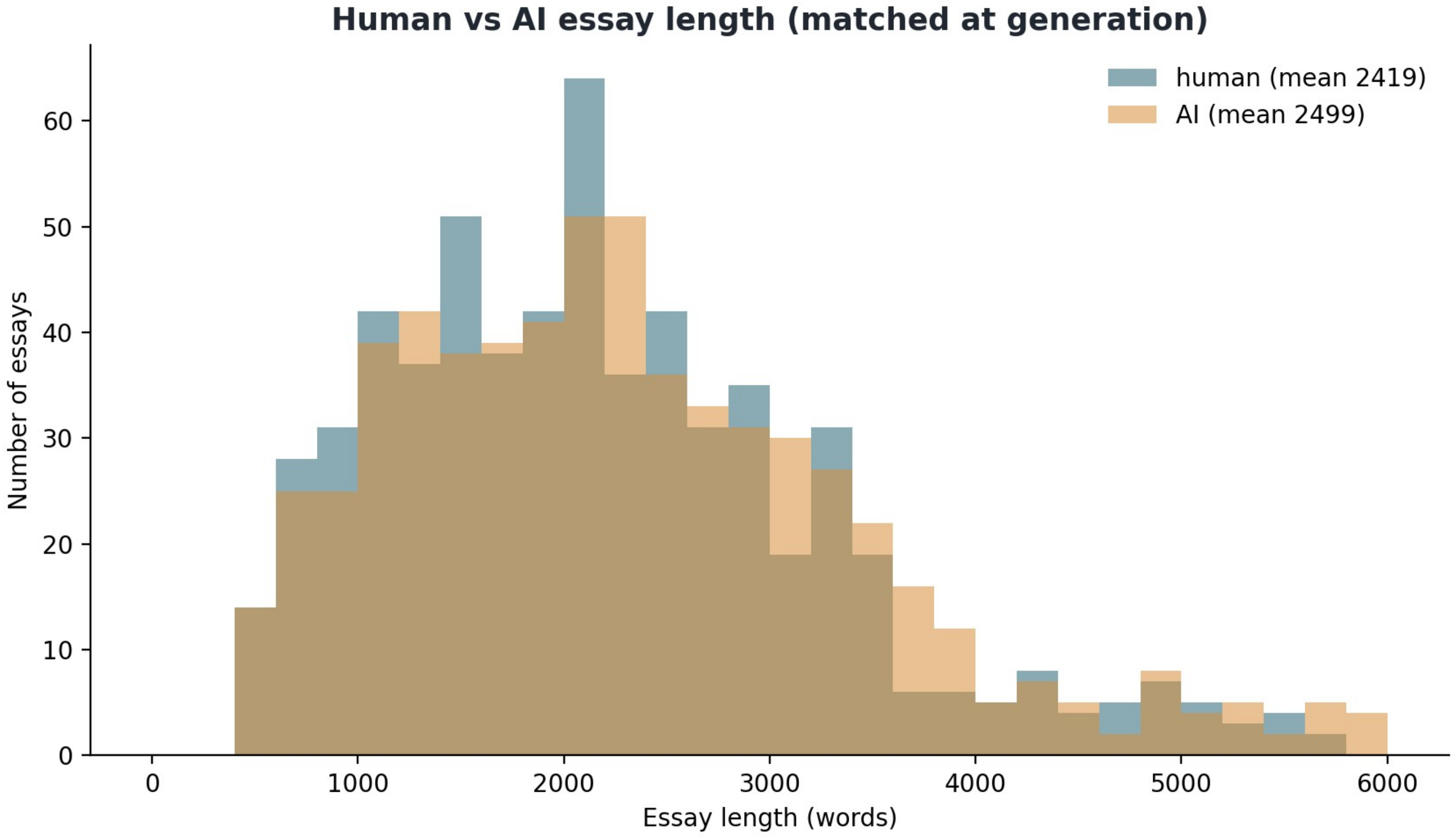


Each step removes a shortcut the detector could cheat with:

balancing removes topic bias, matching removes length, cleaning removes formatting tells, student-level splits remove memorising a writer.

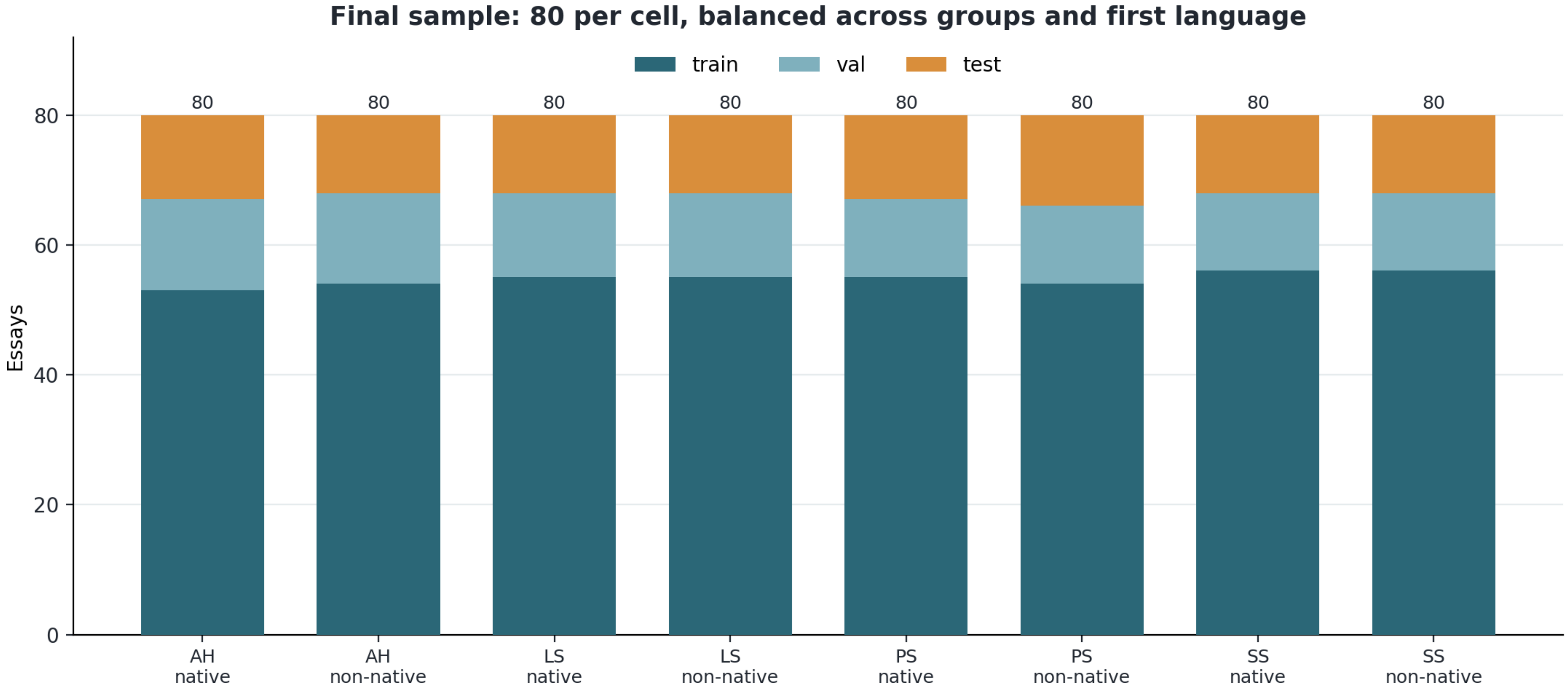
Building block 1: proof the matching worked

The result of the pipeline on the previous slide: 640 real essays and their 640 AI twins. The two length curves sit on top of each other, so the detector cannot cheat by simply noticing that AI essays are longer or shorter. It has to learn writing style.



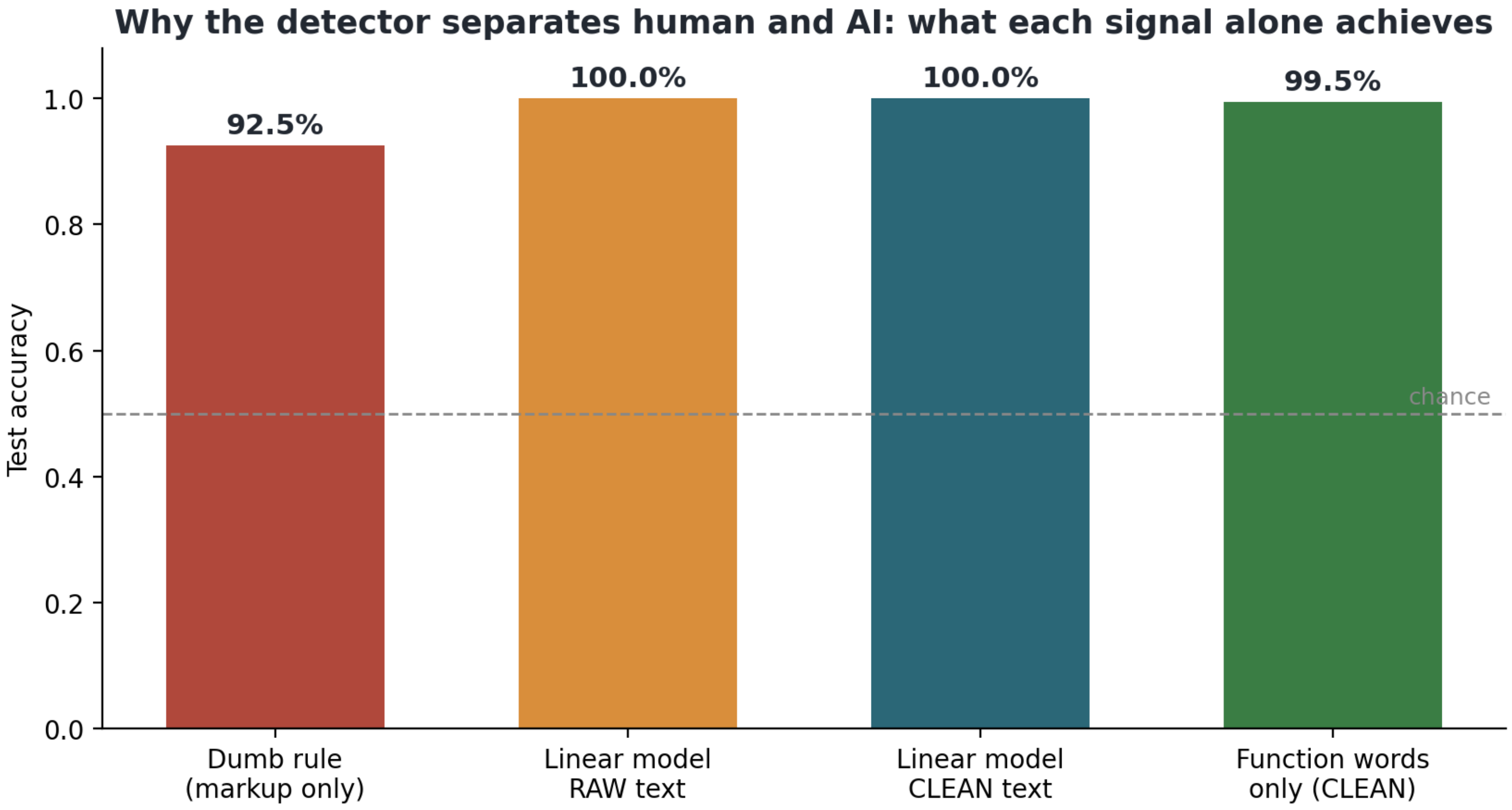
The sample, cell by cell

All eight cells sit at exactly 80 essays: four discipline groups (AH Arts and Humanities, LS Life Sciences, PS Physical Sciences, SS Social Sciences), each split evenly between native and non-native writers, and each bar divided into train, validation and test by student. The detector never gets a reason to prefer a discipline or a language background.



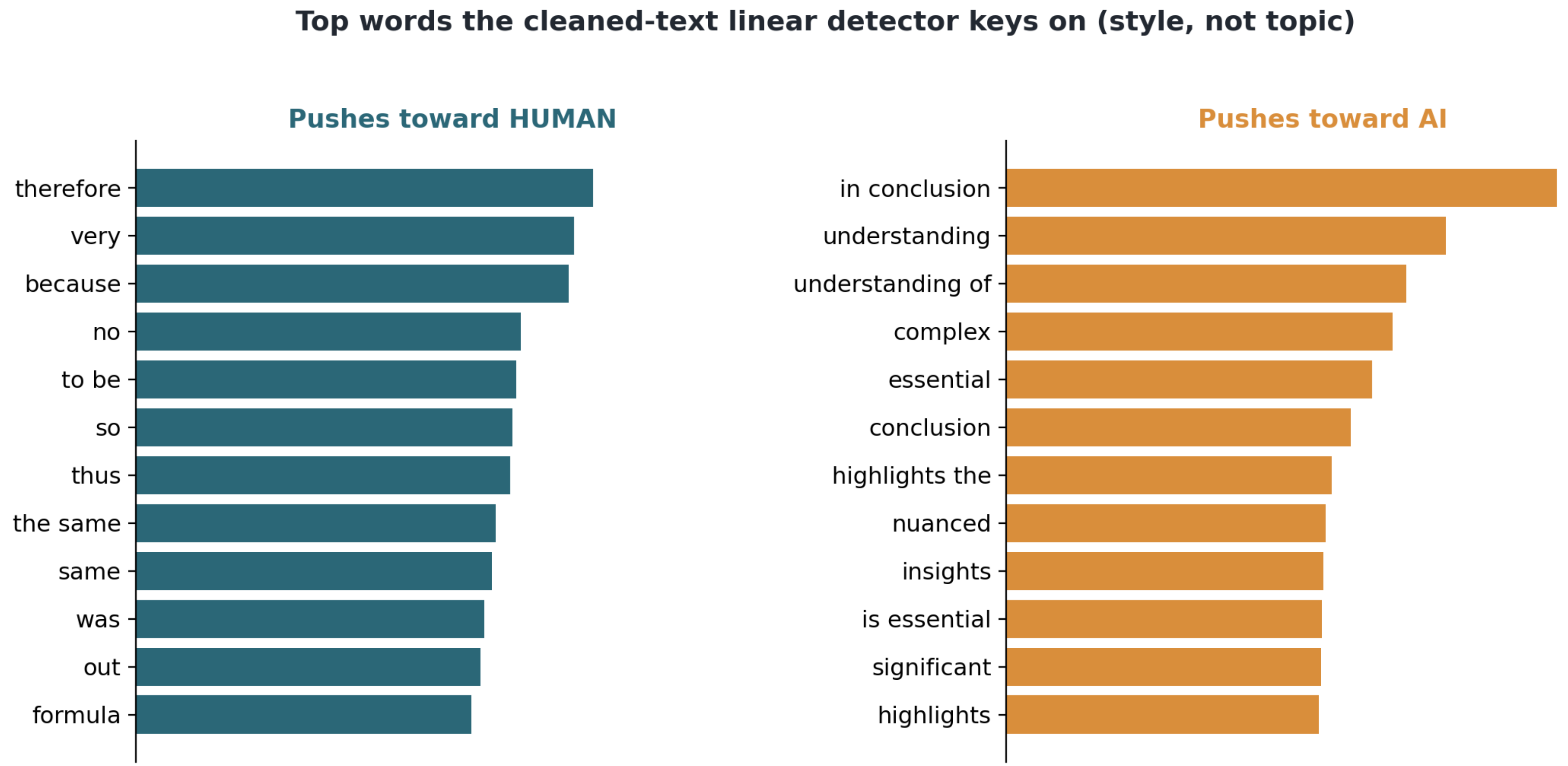
Building block 1: I did not trust a perfect score

My first detector scored a perfect 100%. That is a warning, not a triumph. It turned out the real essays carried tiny hidden formatting tags (invisible leftovers from how the files were saved) the AI ones lacked, nothing to do with writing. I removed it and retrained; the honest score settled at about 99% on essays like the ones it learned from, now genuinely reading writing style.



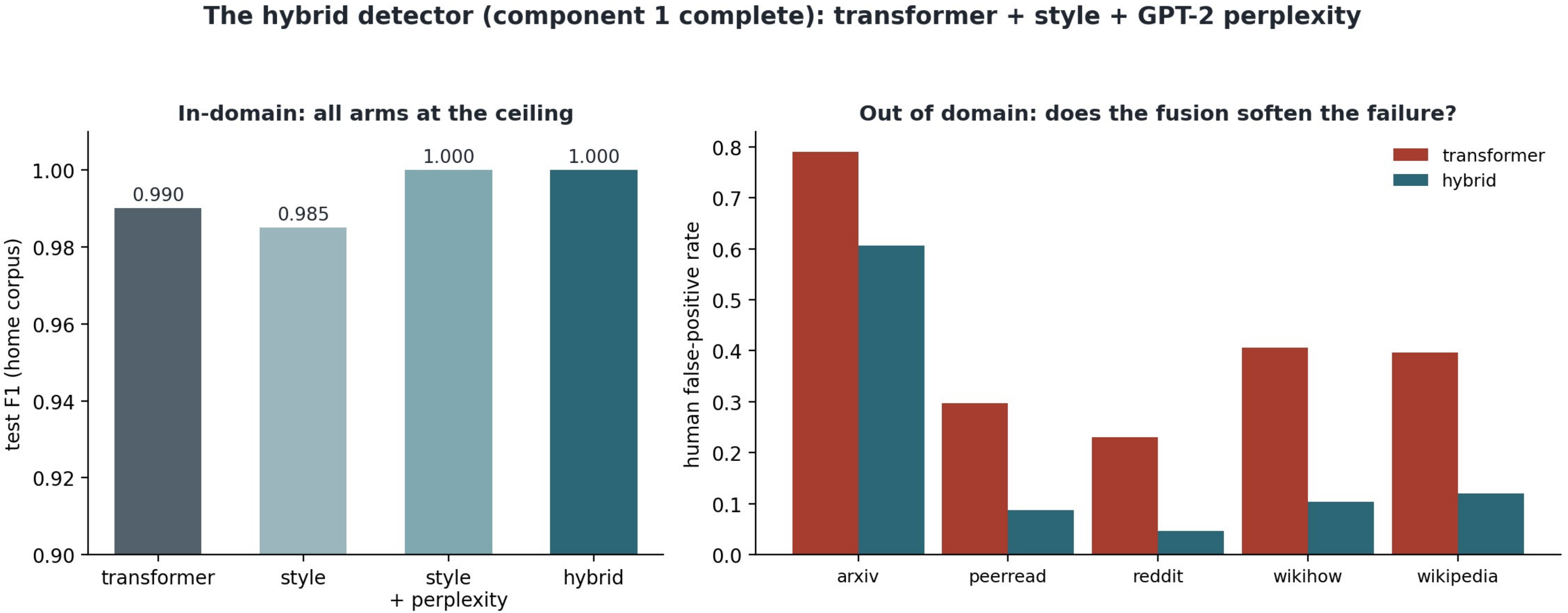
After cleaning: the words that actually separate them

The real top-weighted words of the cleaned-text detector. Students lean on 'therefore', 'very', 'because', 'thus'. The AI leans on 'in conclusion', 'understanding', 'complex', 'nuanced', 'highlights', 'essential'. Style habits, not topics, and you may recognise the AI's favourites.



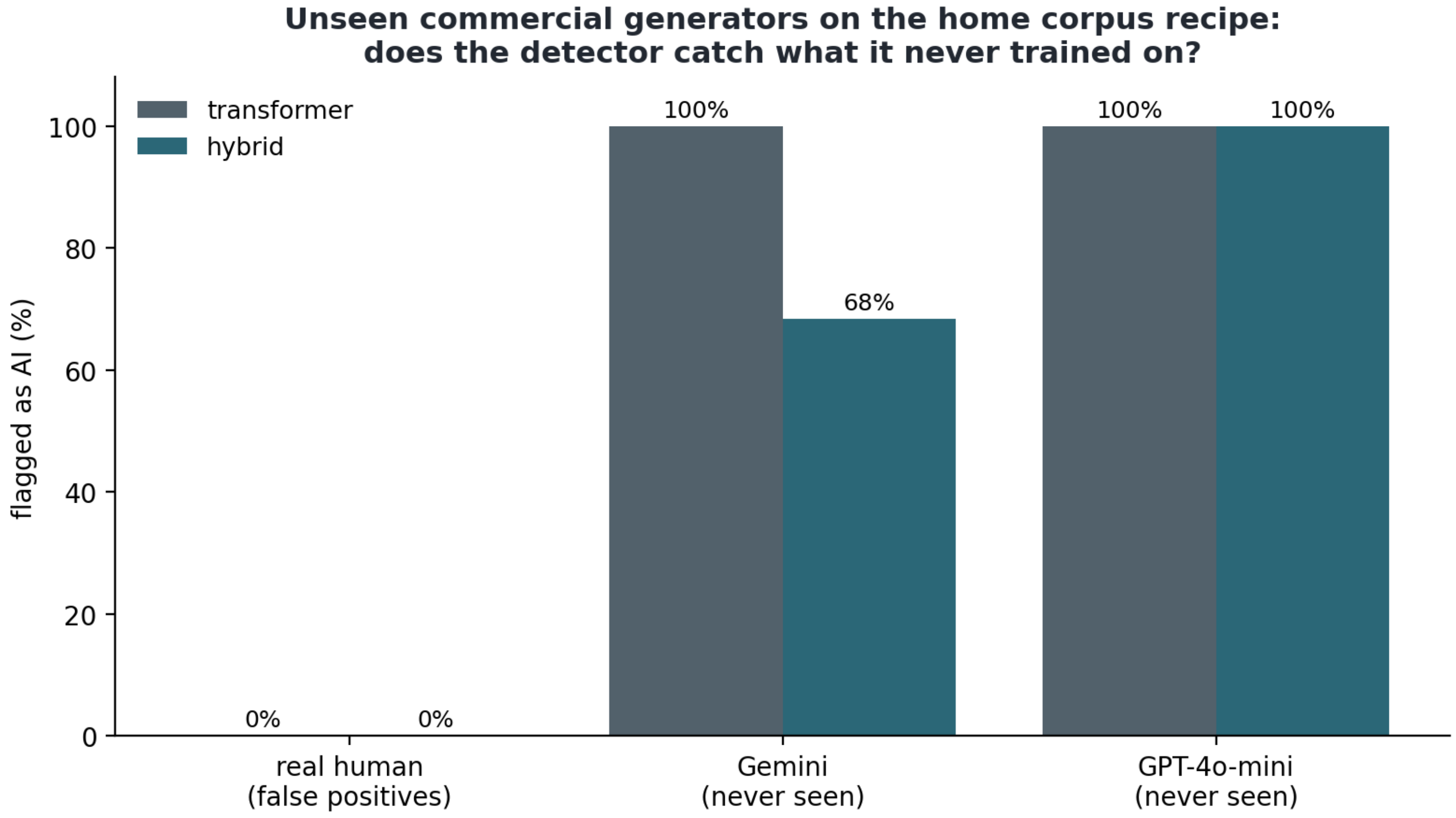
Building block 1: made fairer, on purpose

The finished detector combines two ways of reading an essay: the exact words and phrases chosen, and the shape of the writing (sentence length, how varied it is), like checking both what someone says and how they say it. On its home essays both are near-perfect (left). It also caught 100% of test essays written by two commercial AI models it never saw in training, with zero false accusations of the humans. The hardest test is human writing from other worlds, like forum posts and short science summaries (right): the plain version wrongly flags up to 79% of them as AI, and the combined version cuts those false accusations three to eight times (science abstracts 79% down to about 61%). Fewer wrong accusations, the failure that matters most.



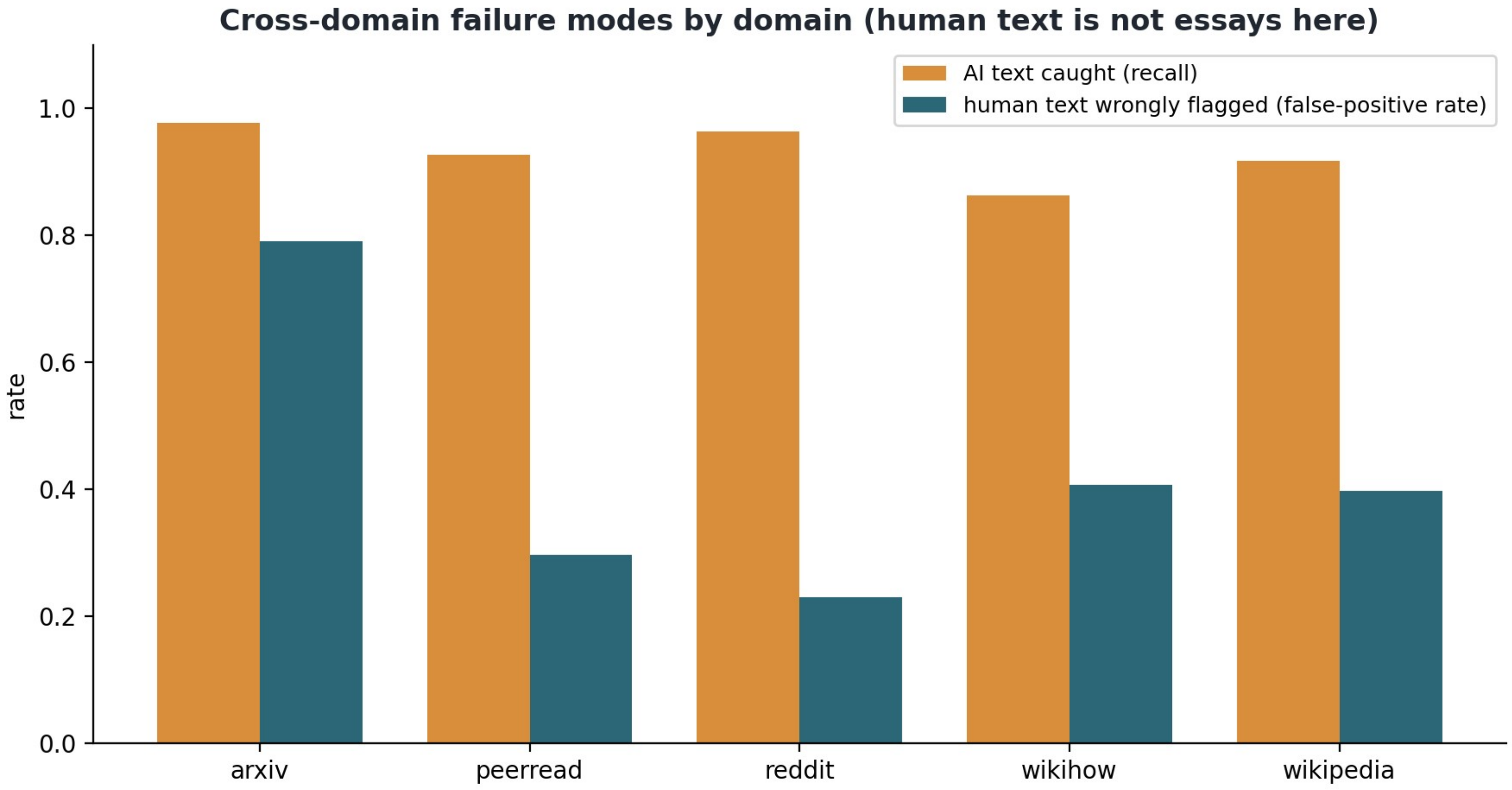
AI models it never saw: every essay caught

Essays generated for the same corpus recipe by two commercial models the detector never trained on. The word-pattern detector (grey) catches 100% from both and falsely flags 0% of the matched human essays. The style-fused hybrid (teal) trades some catch rate on one generator for its cross-domain fairness, a trade-off measured, not hidden.



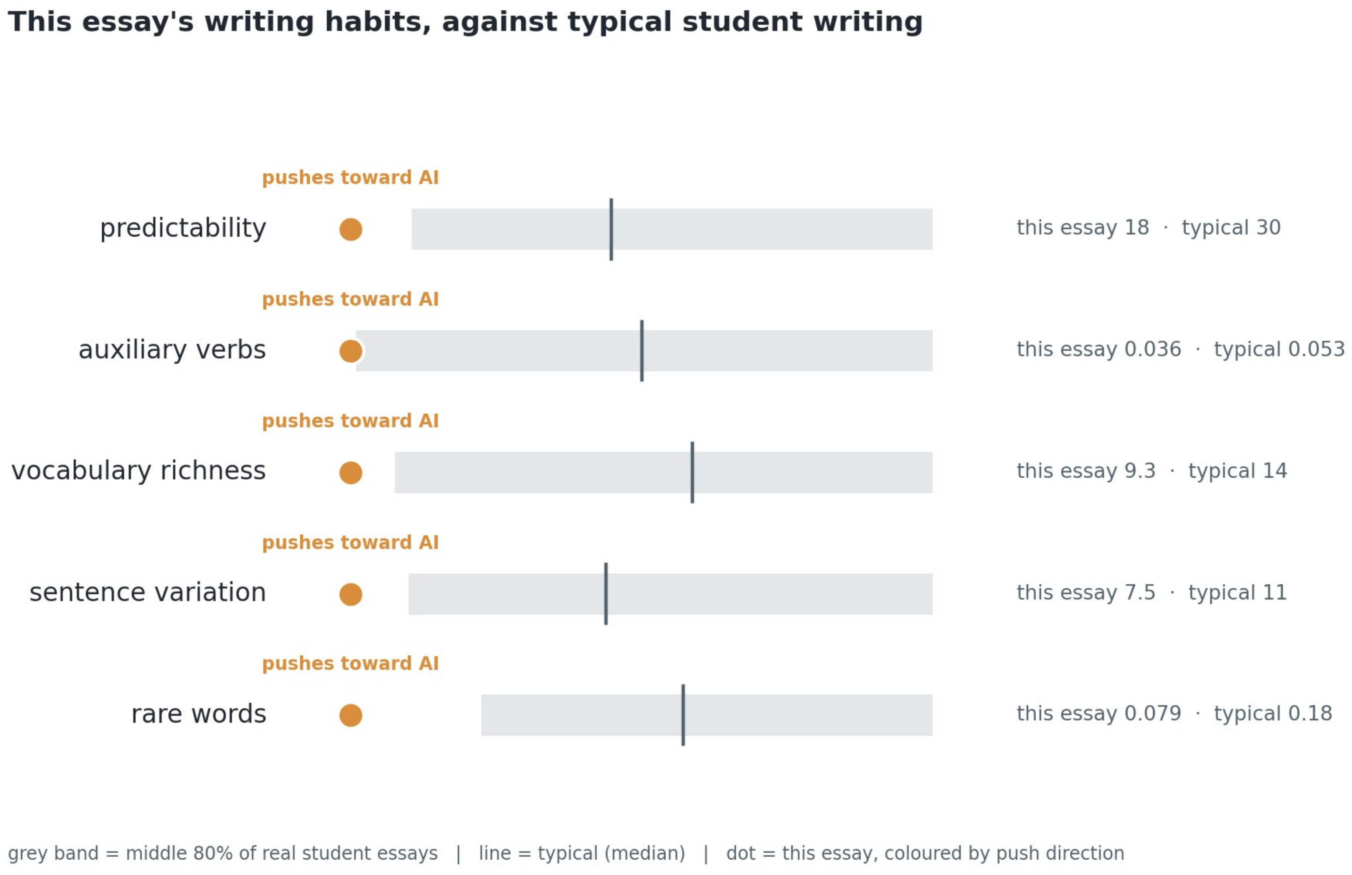
The stress test: human writing from other worlds

An external benchmark of writing the detector never trained on (scientific abstracts, peer reviews, forum posts, wiki articles). Orange: AI text is still caught, 86 to 98%. Teal: real human text wrongly flagged, up to 79% on scientific abstracts. This is the failure mode the hybrid cuts 3 to 8 times, and the reason a flag is never proof.



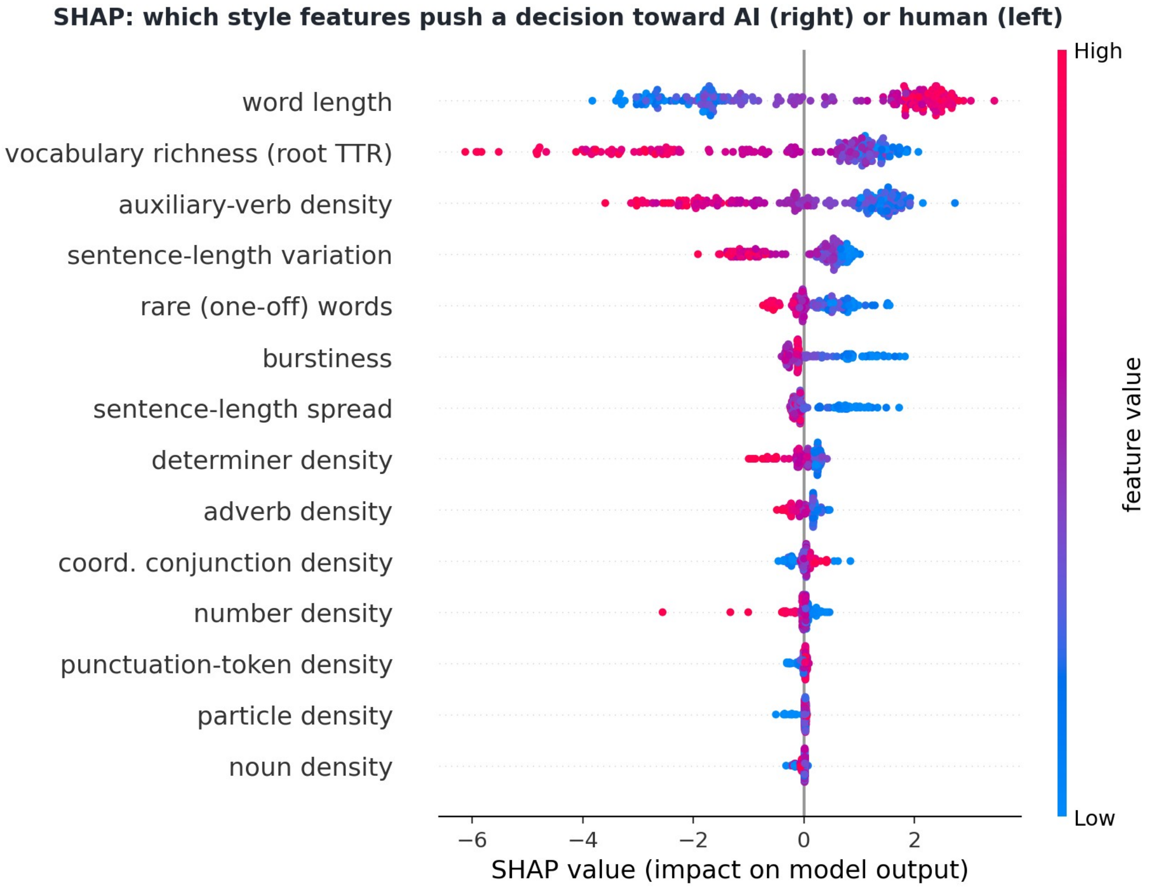
Building block 2: explaining the flag (and where it is still weak)

Instead of a bare number, the lecturer gets this card, straight from the guide for a real flagged essay. Each row is one measurable writing habit. The grey band is the middle 80% of real student essays; the dot is this essay. Every dot sits outside the band: this essay is unusual in five separate ways, and each one is named in plain words. I also test that these reasons are real, not a plausible story. Honest gap: the card has not yet been tested on actual lecturers, and that test is a main job for the second half.



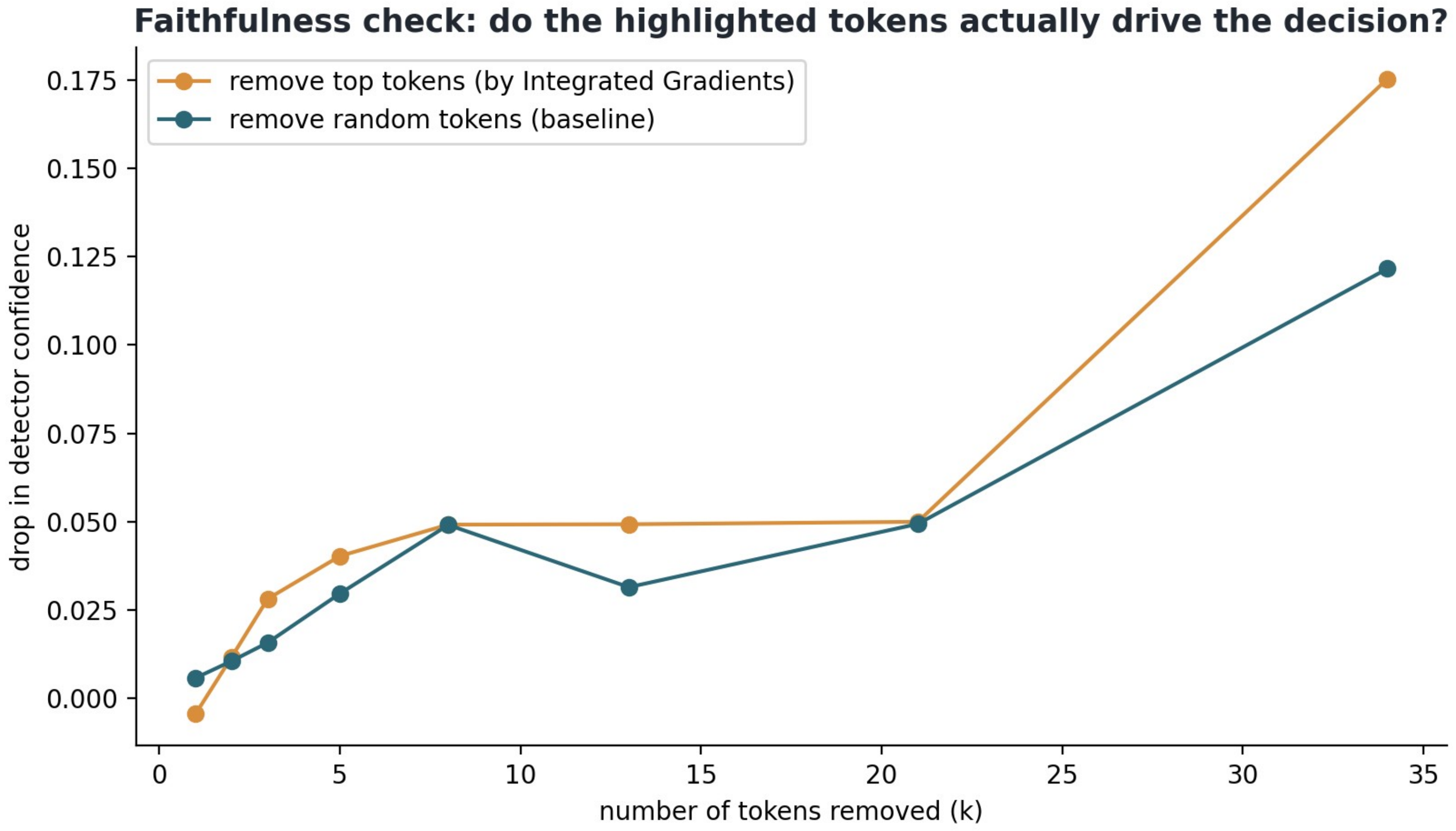
Under the hood of the card

The researcher's view the lecturer card is built from: every dot is one essay, each row one writing habit, and the spread shows how strongly each habit separates real from AI essays across the whole corpus. Word length, vocabulary richness and sentence variation do most of the work. The card is this chart translated into plain language.



Why the card uses habits, not highlighted words

The honesty test behind that choice: delete the words a word-level explanation says matter most (orange) and compare with deleting random words (teal). The lines nearly overlap, so word highlights barely beat chance here; the style signal is spread across the whole essay. The lecturer therefore sees habit-level explanations, which pass this kind of test, not highlighted words, which do not.



Building blocks 3 and 4: from a flag to real questions

For a flagged essay the system finds the student's own claims and, for each, writes questions the true author could answer but a copier could not. Every question traces back to the student's exact sentences, so nothing is invented. Each is tagged by how much thinking it needs (Bloom's taxonomy: Anderson and Krathwohl, 2001).

A claim the system found in the essay: the three set texts belong to different genres but share common themes, and the student argues this shows something about how literature works.

Q1. How did you decide which themes the three texts actually share, and what in each text convinced you?

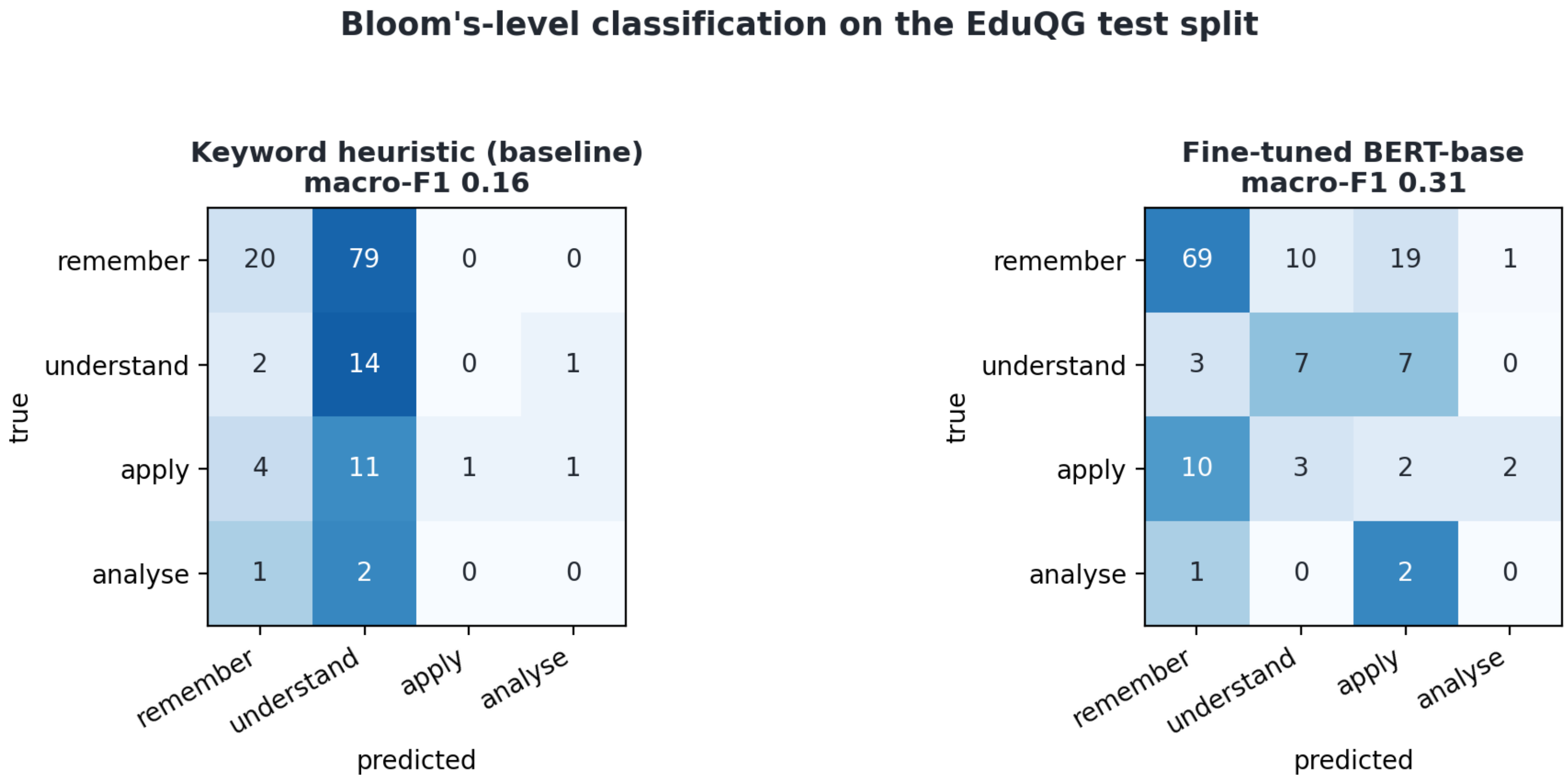
Thinking level: understand

Q2. How does that shared-theme point connect to the bigger argument of your essay?

Thinking level: analyse

Checking the thinking-level labels

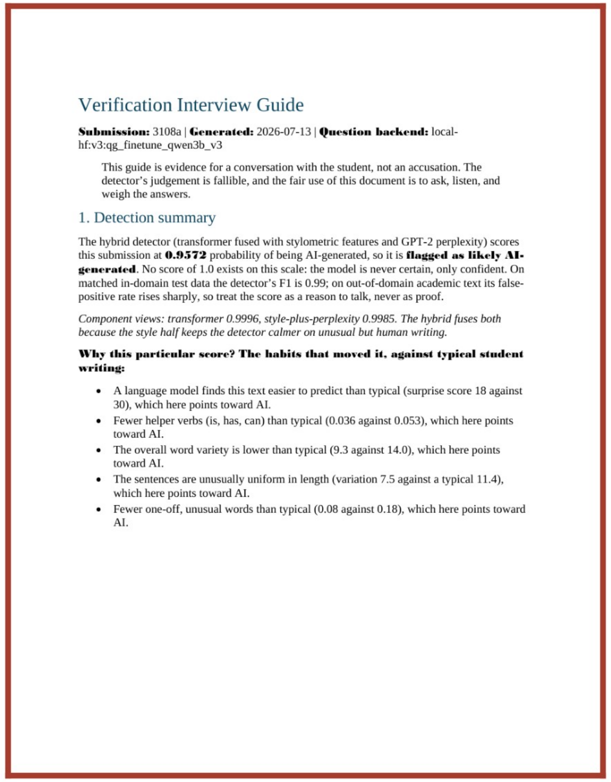
The thinking-level label on each question comes from a trained model, checked against 903 human-labelled questions. It doubles the transparent keyword baseline (macro-F1 0.31 vs 0.16), but the two highest levels have only 110 and 19 labelled examples to learn from, so their tags are marked advisory in the guide. The system states what it can and cannot certify.



The finished product, for both outcomes

Page one of two real guides the system generated for the same essay topic. Left: the AI-written version is flagged at 0.96, the reasons are listed in plain sentences, and the guide goes on to the verification questions. Right: the real student's essay is not flagged at 0.02, and the guide says plainly that no interview is suggested. Same pipeline, both outcomes, nothing invented.

The AI-written version
FLAGGED (score 0.96)



Verification Interview Guide

Submission: 3108a | **Generated:** 2026-07-13 | **Question backend:** local-hf:v3:qg_finetune_qwen3b_v3

> This guide is evidence for a conversation with the student, not an accusation. The detector’s judgement is fallible, and the fair use of this document is to ask, listen, and weigh the answers.

1. Detection summary

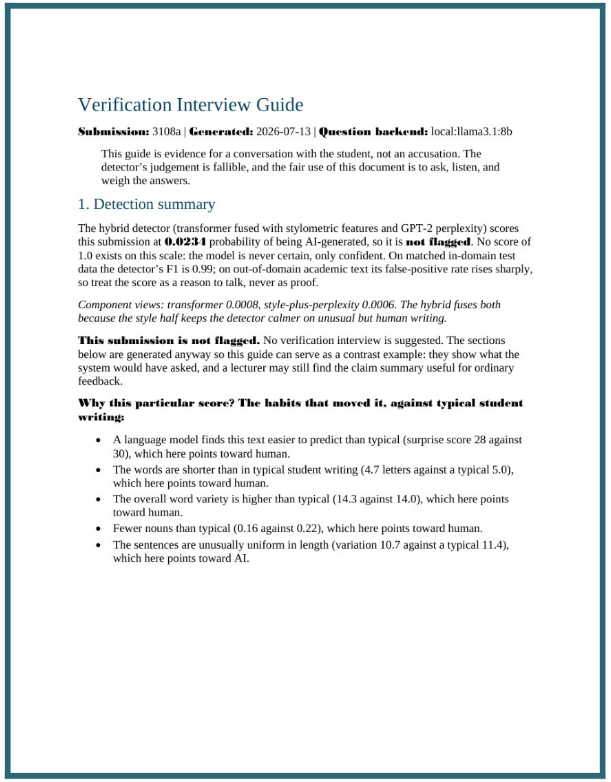
The hybrid detector (transformer fused with stylometric features and GPT-2 perplexity) scores this submission at **0.9572** probability of being AI-generated, so it is **flagged as likely AI-generated**. No score of 1.0 exists on this scale: the model is never certain, only confident. On matched in-domain test data the detector’s F1 is 0.99; on out-of-domain academic text its false-positive rate rises sharply, so treat the score as a reason to talk, never as proof.

Component views: transformer 0.9996, style-plus-perplexity 0.9985. The hybrid fuses both because the style half keeps the detector calmer on unusual but human writing.

Why this particular score? The habits that moved it, against typical student writing:

- A language model finds this text easier to predict than typical (surprise score 18 against 30), which here points toward AI.
- Fewer helper verbs (is, has, can) than typical (0.036 against 0.053), which here points toward AI.
- The overall word variety is lower than typical (9.3 against 14.0), which here points toward AI.
- The sentences are unusually uniform in length (variation 7.5 against a typical 11.4), which here points toward AI.
- Fewer one-off, unusual words than typical (0.08 against 0.18), which here points toward AI.

The real student's essay
NOT FLAGGED (score 0.02)



Verification Interview Guide

Submission: 3108a | **Generated:** 2026-07-13 | **Question backend:** local:llama3.1:8b

> This guide is evidence for a conversation with the student, not an accusation. The detector’s judgement is fallible, and the fair use of this document is to ask, listen, and weigh the answers.

1. Detection summary

The hybrid detector (transformer fused with stylometric features and GPT-2 perplexity) scores this submission at **0.0234** probability of being AI-generated, so it is **not flagged**. No score of 1.0 exists on this scale: the model is never certain, only confident. On matched in-domain test data the detector’s F1 is 0.99; on out-of-domain academic text its false-positive rate rises sharply, so treat the score as a reason to talk, never as proof.

Component views: transformer 0.0008, style-plus-perplexity 0.0006. The hybrid fuses both because the style half keeps the detector calmer on unusual but human writing.

This submission is not flagged. No verification interview is suggested. The sections below are generated anyway so this guide can serve as a contrast example: they show what the system would have asked, and a lecturer may still find the claim summary useful for ordinary feedback.

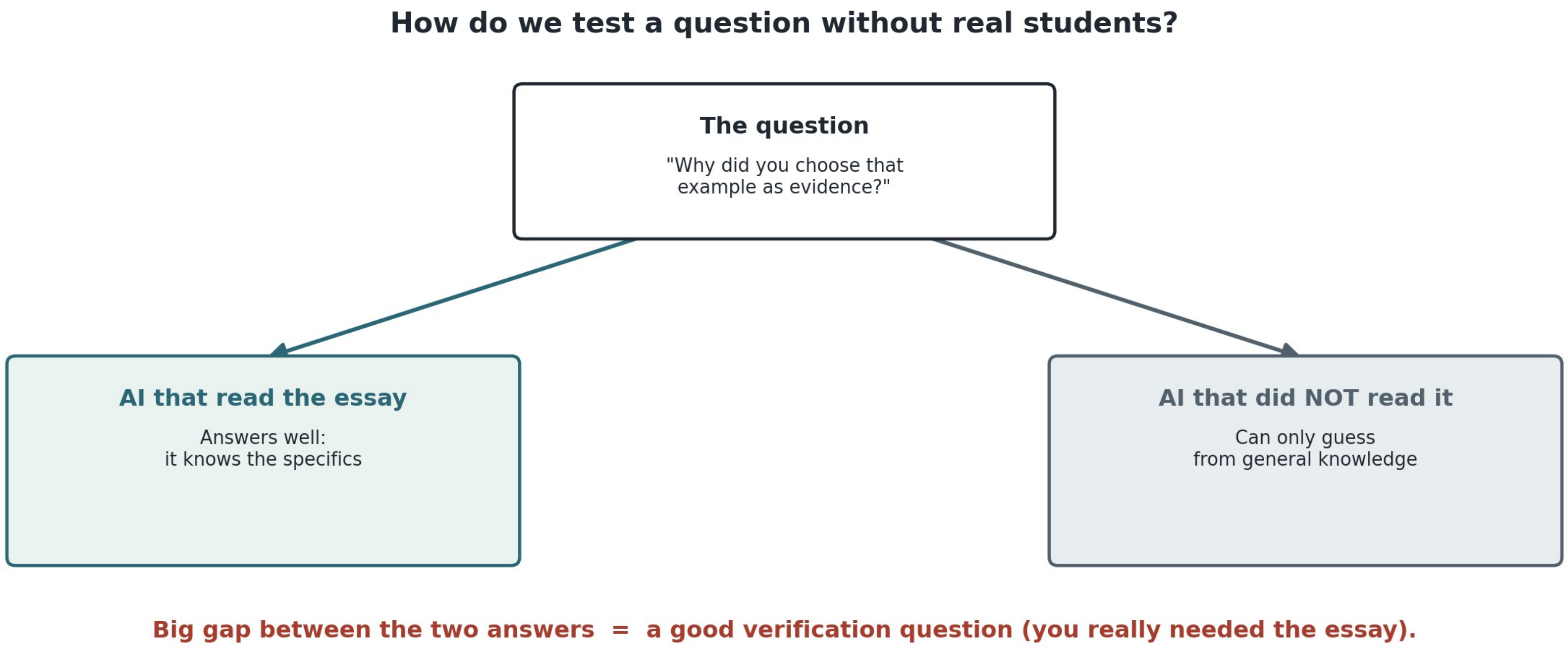
Why this particular score? The habits that moved it, against typical student writing:

- A language model finds this text easier to predict than typical (surprise score 28 against 30), which here points toward human.
- The words are shorter than in typical student writing (4.7 letters against a typical 5.0), which here points toward human.
- The overall word variety is higher than typical (14.3 against 14.0), which here points toward human.
- Fewer nouns than typical (0.16 against 0.22), which here points toward human.
- The sentences are unusually uniform in length (variation 10.7 against a typical 11.4), which here points toward AI.

Both pages are real output from the system, generated for the same essay topic: one guide opens a verification conversation, the other says no interview is needed.

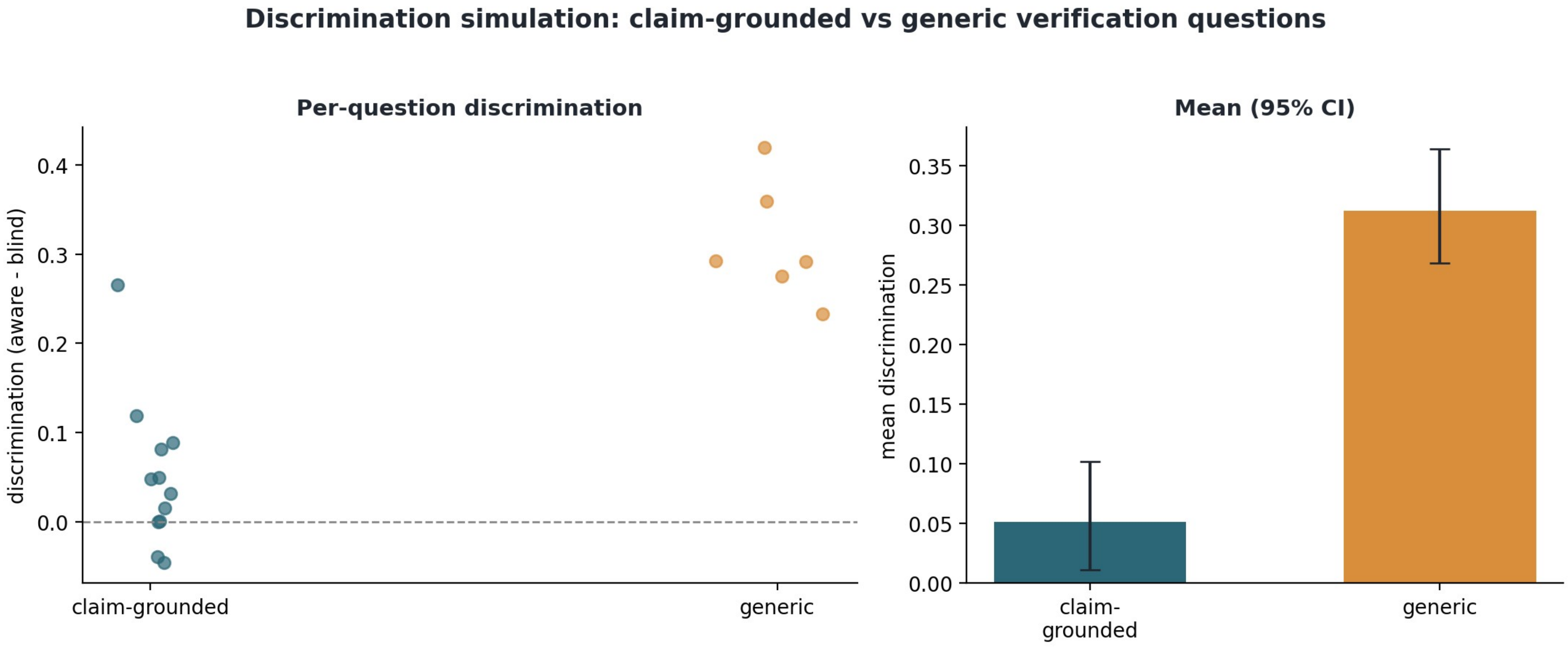
How do you grade a question with no students yet?

A clever trick to test a question without a classroom. Show the question to two AIs: one that has read the essay and one that has not. If the one who read it answers much better, the question genuinely needs the essay, so it is a good verification question. A small gap means anyone could answer, so it is a weak one.



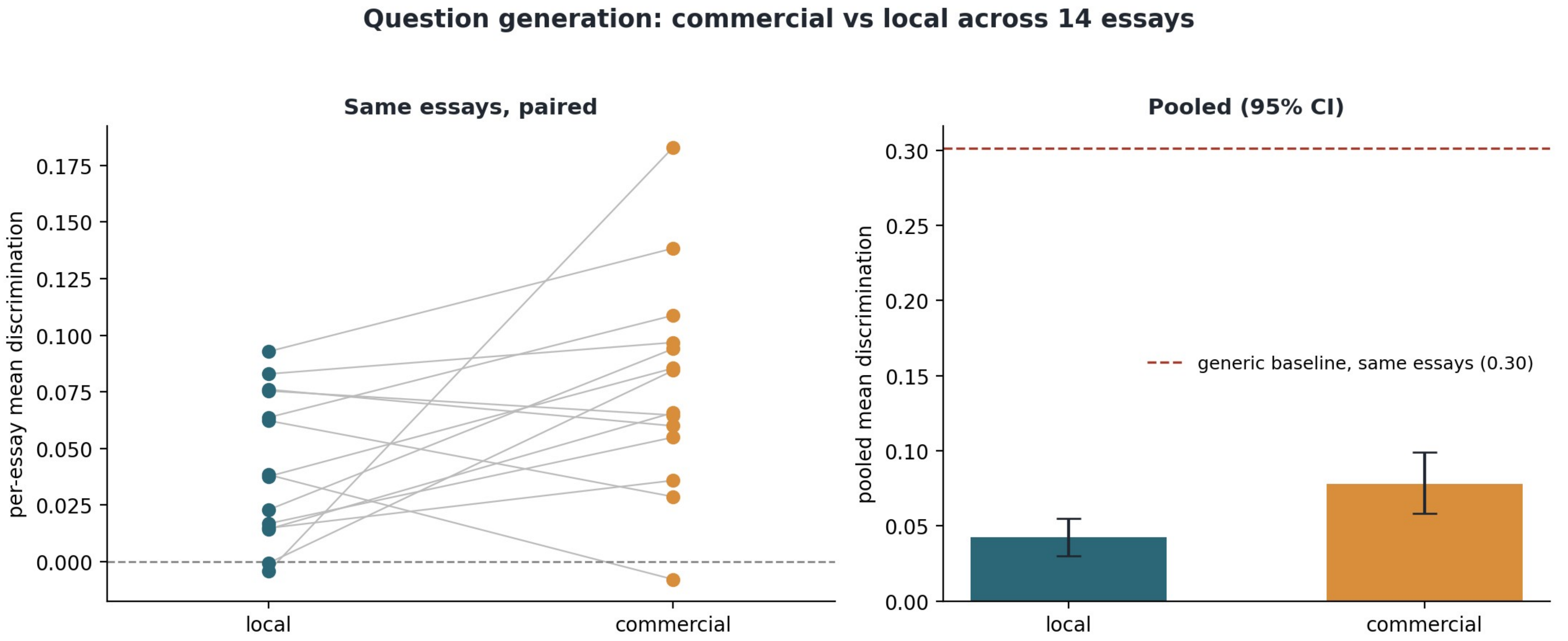
A surprise that changed how I write questions

The trick immediately surprised me. Specific factual questions ('name the feature that...') scored low, because a knowledgeable AI answers them without the essay. Broad questions that force the student to walk through their OWN argument scored far higher. So verification questions should ask for the student's reasoning, not facts anyone could recite.



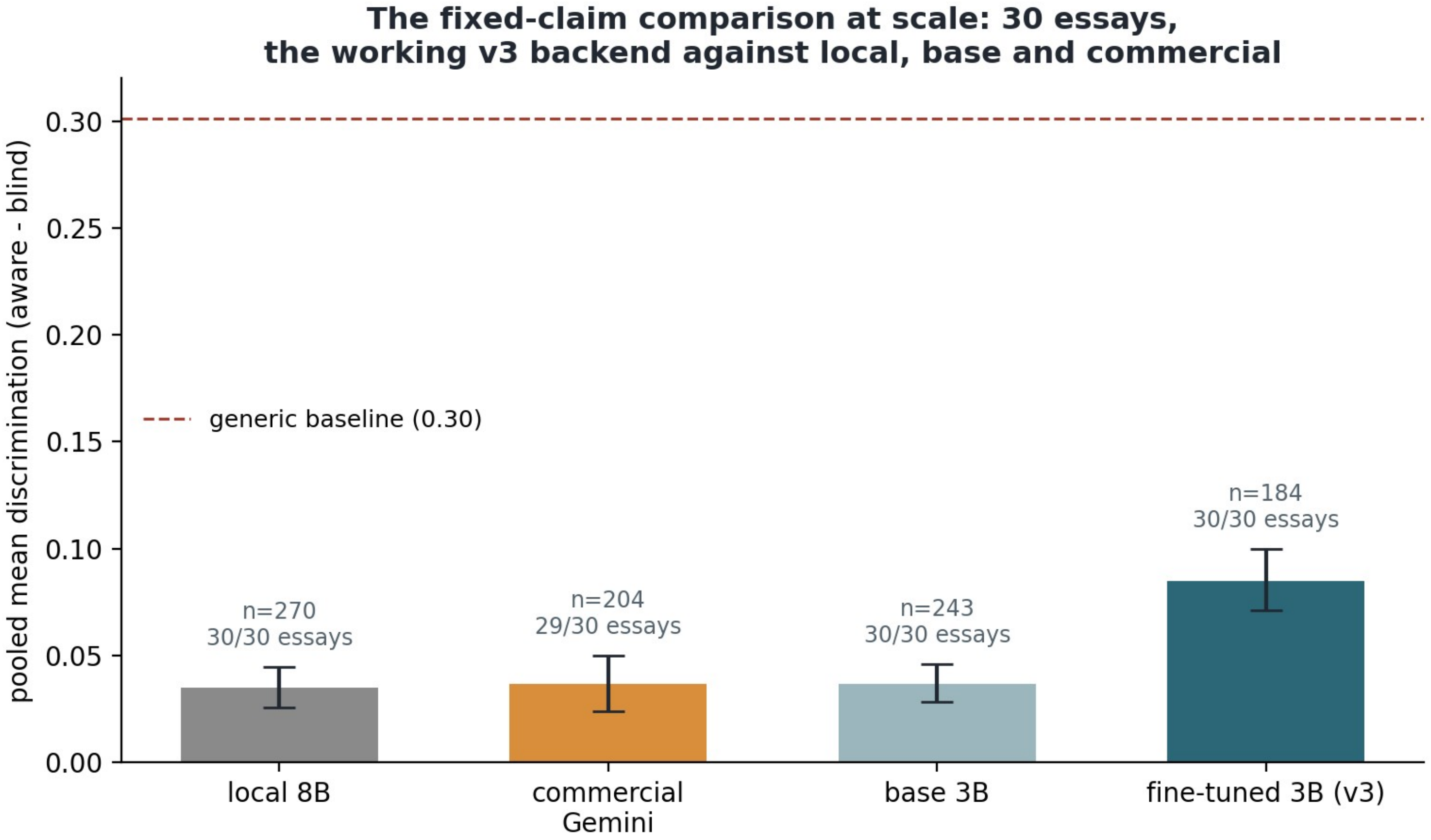
The first race: the cloud model looked better

The first commercial-versus-local comparison, before any fine-tuning: fourteen essays, both backends writing questions for the same essays, one scorer. The cloud model led (0.078 against 0.042, small but real), and both sat far below the generic-question baseline (dashed). This is the lead the fixed-claims experiment on the next slide takes apart.



Laptop model vs expensive cloud model: a fair race

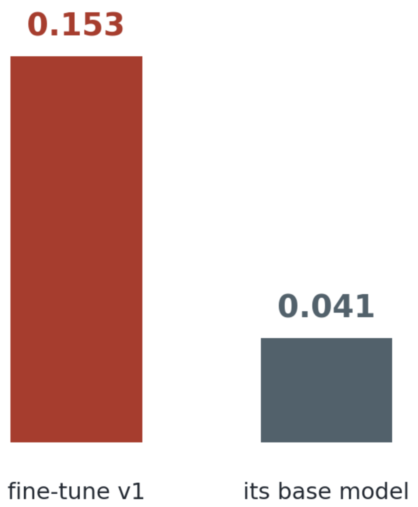
Can a model on my own laptop match a paid cloud model? Normally each model picks which of the student's claims to ask about, and then the cloud model looked a bit better. But when every model must write about exactly the SAME claims (here across 30 essays and 901 questions), the plain models tie, and my fine-tuned laptop model wins: better questions than the cloud model on 24 of the 29 essays both covered, a gap far too consistent to be luck. The earlier cloud lead was about picking easier claims, not writing better questions.



The result that lied to me (my favourite finding)

The left side is what the score claimed: my fine-tuned model looked four times better than its base. The right side is what reading the output revealed: 95% empty question shells that trick the two-AI test, because with nothing real to answer both AIs just guess and the gap looks big for the wrong reason. I proved the cause was the training material and fixed it twice; the bottom-right question is the same model after the fix. The ending came a week later: at thirty essays the corrected model beats the cloud model fairly, with every question well-formed, and that claim stands because this time the quality and the number agree.

The score said



"the fine-tune is 4x better!"

The text said

"Which of the following is correct?"

95% of the fine-tune's output: no options, nothing to answer.
This empty stem alone scores 0.44, above every real question writer.

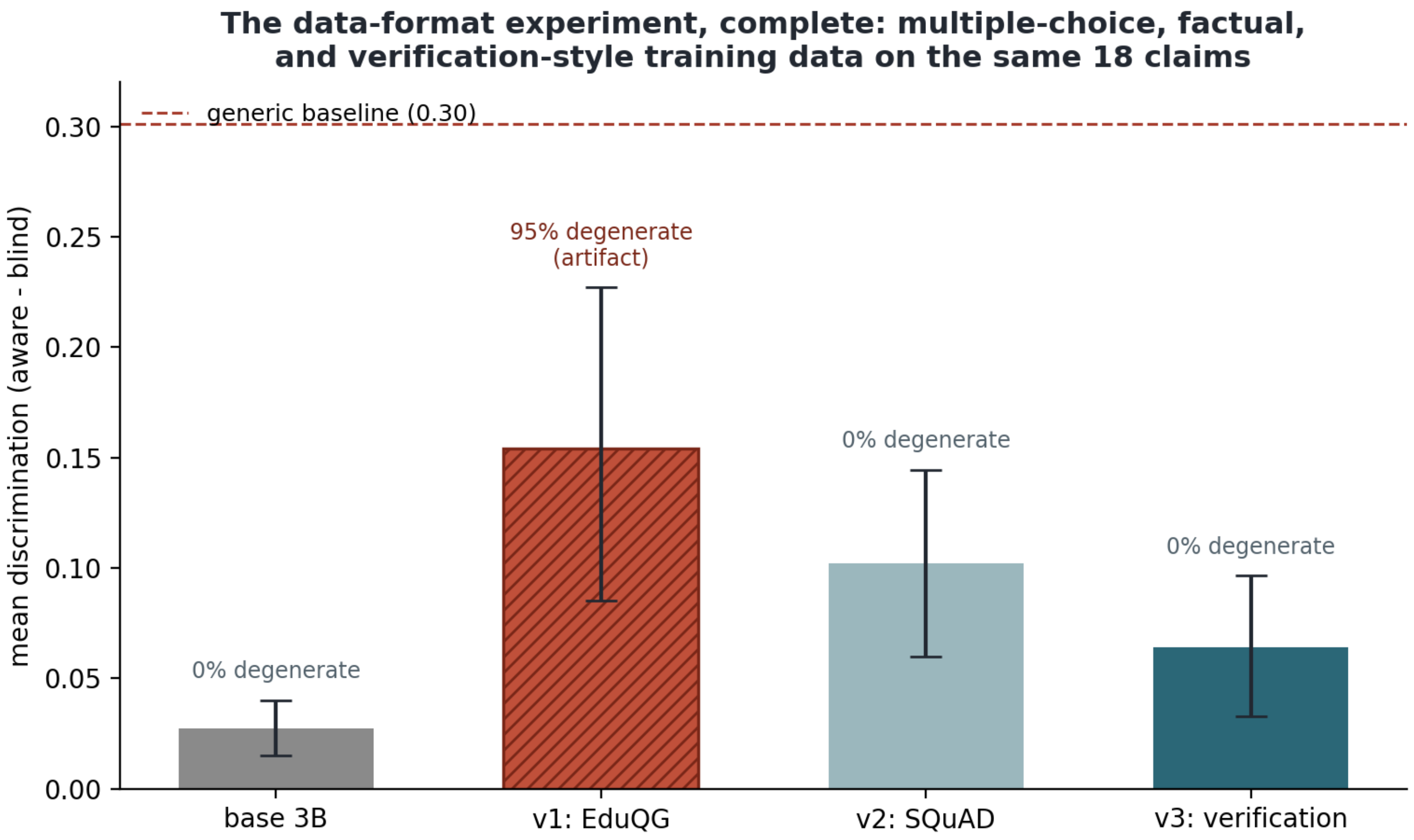
"How did you decide to use the phrase 'challenging decisions made by public authorities' rather than just 'challenging decisions'?"

after retraining on the right data: a real question about the student's own choice

The high score was measuring emptiness. Reading the questions caught it; no number in this project is trusted until the text behind it has been read.

The full data-format experiment, measured

The same model fine-tuned three times, changing only the training data, all scored on the same 18 claims. Multiple-choice data (v1, red) produces the inflated artifact score and 95% broken output. Factual data (v2) and verification-style data (v3) both produce 100% well-formed questions; v2 scores higher but writes the wrong kind of question, so v3 is the one the pipeline ships. The dashed line is the generic-question baseline nobody clears yet.



Where the project is weak right now (and how I will fix it)

- Explainability is not clear enough. It convinces me, not yet a non-technical lecturer. Fix: plain visuals and plain words, tested on real readers. This is the biggest job.
- The questions have never met a real student. My two-AI trick is a stand-in. Fix: a small human study (the guide is ready to be that test).
- The training data is narrow: one source of human essays, one AI model. I have already added test essays from two more AI models (all caught). Fix: widen the training side the same way.
- Every question writer, mine included, still scores below a simple generic-question baseline on my quality measure, and the detector loses accuracy on kinds of writing it was not trained on. Fix: better question training data, and safeguards on unfamiliar writing.
- Samples started small. The main comparison now runs at 30 essays and 901 questions, and the judge check at 60 questions. Fix: keep scaling now that everything runs automatically.

So where am I? About half way

Every building block now exists and the whole pipeline runs end to end, which was the hard part. But the second half is real work, not polishing: making the explanations genuinely understandable, testing on real answers, widening the data, strengthening the weak spots, and the final write-up. I am at the midpoint, with the more visible, human-facing half still ahead.

we are here

BUILT SO FAR (June to mid-July) | **SECOND HALF (mid-July to end of August)**

Done

- ✓ Matched essay dataset (1,280)
- ✓ Hybrid AI-text detector
- ✓ Explainability + honesty test
- ✓ Claim extractor + Bloom classifier
- ✓ Question generation, both backends
- ✓ The evaluation and the fine-tune story
- ✓ The lecturer's guide, end to end

Still to do

- Make the explanations clear and visual
- Validate with real student answers
- Widen the data (more AI models, domains)
- Push the local model further
- Fix the weak spots (see next slide)
- Discussion, polish, submit

The plan for the rest of the summer, and the writing so far

- Now to end July: make the explanations clear and visual; integrate the trained components into the guide (done this week); scale the comparisons; make the detector more careful on unfamiliar kinds of writing.
- August: a small validation of the questions on real answers; widen the data with more AI models; final experiments.
- Late August: protected writing and revision. Hard rule: if time runs short, I cut analysis depth, never the write-up.
- Writing is kept in step with the work: an 88-page draft already exists, twelve chapters including Discussion and Conclusions, with 33 references each checked against the real paper (essential for a project about fake text).
- No human-participant ethics issue for the core work; all data is public and licensed; the output is always evidence for a conversation, never an accusation.

One sentence to take away

- Do not accuse students with a number no one can defend. Detect fairly, explain honestly, and turn a flag into a fair conversation built from the student's own work.
- The method that runs through everything: never trust a score you cannot explain, and never trust one you have not tried to break. It caught three of my own results.
- Status: every piece built and running (detection at 99% with zero false accusations on unseen AI models, laptop question-writer beating the cloud model on a fair test), at the halfway point with the human-facing half still to do.

References and tools

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Models and tools: Llama 3.1 (Grattafiori et al., 2024), Qwen2.5 3B (Yang et al., 2024), Gemini (Google), Nomic Embed (Nussbaum et al., 2024).

Full reference list (33 entries, each verified against the primary source) in the dissertation draft.